

Laplace Transform Tables

Specific Unilateral Laplace Transform Pairs			
Signal	$x(t) = \mathcal{L}^{-1}[X(s)]$		$X(s) = \int_0^\infty x(t)e^{-st}dt$
Unit impulse	$\delta(t)$	$\Longleftrightarrow$	1
Unit step	$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$	$\Longleftrightarrow$	$\frac{1}{s}$
Exponential	$e^{-\alpha t} u(t)$	$\Longleftrightarrow$	$\frac{1}{s + \alpha}$
Power of t	$t^n u(t)$	$\Longleftrightarrow$	$\frac{n!}{s^{n+1}}$
Damped power of t	$t^n e^{-\alpha t} u(t)$	$\Longleftrightarrow$	$\frac{n!}{(s + \alpha)^{n+1}}$
Sine	$\sin(\beta t) u(t)$	$\Longleftrightarrow$	$\frac{\beta}{s^2 + \beta^2}$
Cosine	$\cos(\beta t) u(t)$	$\Longleftrightarrow$	$\frac{s}{s^2 + \beta^2}$
Damped sine	$e^{-\alpha t} \sin(\beta t) u(t)$	$\Longleftrightarrow$	$\frac{\beta}{(s + \alpha)^2 + \beta^2}$
Damped cosine	$e^{-\alpha t} \cos(\beta t) u(t)$	$\Longleftrightarrow$	$\frac{s + \alpha}{(s + \alpha)^2 + \beta^2}$
t times damped sine	$t e^{-\alpha t} \sin(\beta t) u(t)$	$\Longleftrightarrow$	$\frac{2\beta (s + \alpha)}{((s + \alpha)^2 + \beta^2)^2}$
t times damped cosine	$t e^{-\alpha t} \cos(\beta t) u(t)$	$\Longleftrightarrow$	$\frac{(s + \alpha)^2 - \beta^2}{((s + \alpha)^2 + \beta^2)^2}$

Unilateral Laplace Transform Properties			
Property	$x(t) = \mathcal{L}^{-1}[X(s)]$		$X(s) = \int_0^\infty x(t)e^{-st} dt$
Linearity	$Ax_1(t) + Bx_2(t)$	$\iff$	$AX_1(s) + BX_2(s)$
Scale change	$x(at)$	$\iff$	$\frac{1}{a} X\left(\frac{s}{a}\right), a > 0$
Time shift	$x(t-T)u(t-T)$	$\iff$	$e^{-sT} X(s), T > 0$
s -shift	$e^{-\alpha t} x(t)$	$\iff$	$X(s+\alpha)$
Differentiation	$\frac{dx(t)}{dt}$	$\iff$	$sX(s) - x(0^-)$
Times t	$tx(t)$	$\iff$	$\frac{-dX(s)}{ds}$
Integration	$\int_0^t x(\tau) d\tau$	$\iff$	$\frac{X(s)}{s}$
Convolution	$\int_0^t x_1(\tau) x_2(t-\tau) d\tau$	$\iff$	$X_1(s) X_2(s)$
Conjugation	$x^*(t)$	$\iff$	$X^*(s^*)$
Initial value (if it exists)	$x(0^+)$	$\iff$	$\lim_{s \rightarrow \infty} sX(s)$
Final value (if it exists)	$x(\infty)$	$\iff$	$\lim_{s \rightarrow 0} sX(s)$

Fourier Transform Tables

Specific Fourier Transform Pairs			
Signal	$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi f t} df$		$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$
Unit Impulse	$\delta(t)$	$\Longleftrightarrow$	1
Constant	1	$\Longleftrightarrow$	$\delta(f)$
Unit Step	$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$	$\Longleftrightarrow$	$\frac{\delta(f)}{2} + \frac{1}{j2\pi f}$
Signum	$\text{sgn}(t) = \begin{cases} 1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$	$\Longleftrightarrow$	$\frac{1}{j\pi f}$
Exponential 1-sided	$e^{-at} u(t)$	$\Longleftrightarrow$	$\frac{1}{a + j2\pi f}$
Gaussian	$e^{-\pi t^2}$	$\Longleftrightarrow$	$e^{-\pi f^2}$
Sine	$\sin(2\pi f_0 t)$	$\Longleftrightarrow$	$\frac{\delta(f-f_0) - \delta(f+f_0)}{2j}$
Cosine	$\cos(2\pi f_0 t)$	$\Longleftrightarrow$	$\frac{\delta(f-f_0) + \delta(f+f_0)}{2}$
Sine ($t > 0$)	$\sin(2\pi f_0 t) u(t)$	$\Longleftrightarrow$	$\frac{\delta(f-f_0) - \delta(f+f_0)}{4j} + \frac{1}{2\pi} \frac{f_0}{f_0^2 - f^2}$
Cosine ($t > 0$)	$\cos(2\pi f_0 t) u(t)$	$\Longleftrightarrow$	$\frac{\delta(f-f_0) + \delta(f+f_0)}{4} + \frac{1}{j2\pi} \frac{f}{f^2 - f_0^2}$
Rectangular	$x(t) = \begin{cases} 1, & t < \tau/2 \\ 0, & \text{otherwise} \end{cases}$	$\Longleftrightarrow$	$\frac{\sin(\pi f \tau)}{\pi f}$
“sin x over x”	$\frac{\sin(2\pi W t)}{\pi t}$	$\Longleftrightarrow$	$X(f) = \begin{cases} 1, & f < W \\ 0, & \text{otherwise} \end{cases}$
Hilbert Transform	$\frac{1}{\pi t}$	$\Longleftrightarrow$	$-j \text{sgn}(f)$
Tone FM	$e^{j\beta \sin(2\pi f_m t)} = \sum_{k=-\infty}^{\infty} J_k(\beta) e^{j2\pi k f_m t}$	$\Longleftrightarrow$	$\sum_{k=-\infty}^{\infty} J_k(\beta) \delta(f - k f_m)$
Pulse Train	$\sum_{m=-\infty}^{\infty} \delta(t - mT)$	$\Longleftrightarrow$	$\frac{1}{T} \sum_{k=-\infty}^{\infty} \delta(f - \frac{k}{T})$

Bessel function, 1'st kind, order k : $J_k(\beta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j(\beta \sin x - kx)} dx$.

Fourier Transform Properties			
Property	$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df$		$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$
Linearity	$Ax_1(t) + Bx_2(t)$	$\iff$	$AX_1(f) + BX_2(f)$
Duality	$X(t)$	$\iff$	$x(-f)$
Scale Change	$x(at), a \neq 0$	$\iff$	$\frac{1}{ a } X\left(\frac{f}{a}\right)$
Time Shift	$x(t - t_0)$	$\iff$	$X(f) e^{-j2\pi ft_0}$
Frequency Shift	$x(t) e^{j2\pi f_0 t}$	$\iff$	$X(f - f_0)$
Convolution	$\int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$	$\iff$	$X(f) H(f)$
Multiplication	$v(t) w(t)$	$\iff$	$\int_{-\infty}^{\infty} V(\mu) W(f - \mu) d\mu$
Differentiation	$\frac{dx(t)}{dt}$	$\iff$	$j2\pi f X(f)$
Times t	$t x(t)$	$\iff$	$\frac{-1}{j2\pi} \frac{dX(f)}{df}$
Integration	$\int_{-\infty}^t x(\tau) d\tau$	$\iff$	$\frac{X(f)}{j2\pi f} + \frac{X(0)}{2} \delta(f)$
Even Symmetry	$x(t) = x(-t)$	$\iff$	$X(f) = X(-f)$
Odd Symmetry	$x(t) = -x(-t)$	$\iff$	$X(f) = -X(-f)$
Real $x(t)$	$x(t) = x^*(t)$	$\iff$	$X(f) = X^*(-f)$ ($ X(f) $ even, $\angle X(f)$ odd)
Periodic $x(t)$ (X_k : FS Coefficients)	$\sum_{k=-\infty}^{\infty} X_k e^{j2\pi kt/T_1}$	$\iff$	$\sum_{k=-\infty}^{\infty} X_k \delta(f - \frac{k}{T_1})$
Parseval	$\int_{-\infty}^{\infty} x(t) ^2 dt$	$=$	$\int_{-\infty}^{\infty} X(f) ^2 df$

Fourier series (period T_1): $X_k = \frac{1}{T_1} \int_{T_1} x(t) e^{-j2\pi kt/T_1} dt \iff x(t) = \sum_{k=-\infty}^{\infty} X_k e^{j2\pi kt/T_1}.$

Fourier Series Tables

Specific Fourier Series Pairs, T_1 : Period		
Signal	$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{j2\pi kt/T_1}$	$X_k = \frac{1}{T_1} \int_{T_1} x(t) e^{-j2\pi kt/T_1} dt$
Unit Impulse	$\delta(t)$	$\Longleftrightarrow \frac{1}{T_1}$
Constant	1	$\Longleftrightarrow \frac{\sin(\pi k)}{\pi k} = \begin{cases} 1, & k = 0 \\ 0, & \text{otherwise} \end{cases}$
Complex Exponential	$e^{\pm j2\pi mt/T_1}$	$\Longleftrightarrow \frac{\sin(\pi(k \mp m))}{\pi(k \mp m)} = \begin{cases} 1, & k = \pm m \\ 0, & \text{otherwise} \end{cases}$
Sine, $f = \frac{m}{T_1}$	$\sin(2\pi mt/T_1)$	$\Longleftrightarrow \begin{cases} 1/(2j), & k = m \\ -1/(2j), & k = -m \\ 0, & \text{otherwise} \end{cases}$
Cosine, $f = \frac{m}{T_1}$	$\cos(2\pi mt/T_1)$	$\Longleftrightarrow \begin{cases} 1/2, & k = m \\ 0, & \text{otherwise} \end{cases}$
Rectangular	$x(t) = \begin{cases} 1, & t < \tau/2 \\ 0, & \text{otherwise} \end{cases}$	$\Longleftrightarrow \frac{\sin(\pi k \tau/T_1)}{\pi k}$

Fourier Series Properties, T_1 : Period		
Property	$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{j2\pi kt/T_1}$	$X_k = \frac{1}{T_1} \int_{T_1} x(t) e^{-j2\pi kt/T_1} dt$
Linearity	$Ax_1(t) + Bx_2(t)$	$\Longleftrightarrow AX_1[k] + BX_2[k]$
Scale Change	$x(at), \quad (x(t) = 0, t \geq \frac{T_1}{a})$	$\Longleftrightarrow \frac{1}{ a } X[k/a], \quad a \neq 0$
Time Shift	$x(t - t_0)$	$\Longleftrightarrow X_k e^{-j2\pi kt_0/T_1}$
Frequency Shift	$x(t) e^{j2\pi k_0 t/T_1}$	$\Longleftrightarrow X_{k-k_0}$
Convolution	$\frac{1}{T_1} \int_{T_1} x(\tau) h(t - \tau) d\tau$	$\Longleftrightarrow X_k H_k$
Multiplication	$v(t) w(t)$	$\Longleftrightarrow \sum_{m=-\infty}^{\infty} V_m W_{k-m}$
Even Symmetry	$x(t) = x(-t)$	$\Longleftrightarrow X_k = X_{-k}$
Odd Symmetry	$x(t) = -x(-t)$	$\Longleftrightarrow X_k = -X_{-k}$
Real $x(t)$	$x(t) = x^*(t)$	$\Longleftrightarrow \begin{aligned} &X_k = X_{-k}^* \\ &(X_k \text{ even}, \angle X_k \text{ odd}) \end{aligned}$
Parseval	$\frac{1}{T_1} \int_{T_1} x(t) ^2 dt$	$= \sum_{k=-\infty}^{\infty} X_k ^2$

Discrete-Time Fourier Transform Tables

Specific Discrete-Time Fourier Transform Pairs			
Signal	$x_n = \int_1 X(\phi) e^{j2\pi\phi n} d\phi$		$X(\phi) = \sum_{n=-\infty}^{\infty} x_n e^{-j2\pi\phi n}$
Unit Impulse	$\delta_n = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$	$\iff$	1
Constant	1	$\iff$	$\sum_{k=-\infty}^{\infty} \delta(\phi - k)$
Unit Step	$u_n = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$	$\iff$	$\frac{1}{1 - e^{-j2\pi\phi}} + \frac{1}{2} \sum_{k=-\infty}^{\infty} \delta(\phi - k)$
Signum	$\text{sgn}(n) = \begin{cases} 1, & n > 0 \\ 0, & n = 0 \\ -1, & n < 0 \end{cases}$	$\iff$	$\frac{1}{j \tan(\pi\phi)}$
Exponential 1-sided	$x_n = \begin{cases} a^n, & n \geq 0, a < 1 \\ 0, & \text{otherwise} \end{cases}$	$\iff$	$\frac{1}{1 - a e^{-j2\pi\phi}}$
Sine	$\sin(2\pi\phi_0 n)$	$\iff$	$\sum_{k=-\infty}^{\infty} \frac{\delta(\phi - \phi_0 - k) - \delta(\phi + \phi_0 - k)}{2j}$
Cosine	$\cos(2\pi\phi_0 n)$	$\iff$	$\sum_{k=-\infty}^{\infty} \frac{\delta(\phi - \phi_0 - k) + \delta(\phi + \phi_0 - k)}{2}$
Rectangular	$x_n = \begin{cases} 1, & n \leq d \\ 0, & \text{otherwise} \end{cases}$	$\iff$	$\frac{\sin(\pi\phi(2d+1))}{\sin(\pi\phi)}$
“sin x over x”	$2B \frac{\sin(2\pi Bn)}{2\pi Bn}$	$\iff$	$X(\phi) = \begin{cases} 1, & \phi - k < B, k \text{ integer} \\ 0, & \text{otherwise} \end{cases}$
Hilbert Transform	$x_n = \begin{cases} 2/(\pi n) & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$	$\iff$	$-j \text{sgn}(\phi), \quad -\frac{1}{2} < \phi < \frac{1}{2}$
Pulse Train	$\sum_{m=-\infty}^{\infty} \delta_{n-mN}$	$\iff$	$\frac{1}{N} \sum_{m=-\infty}^{\infty} \delta(\phi - \frac{m}{N})$

Discrete-Time Fourier Transform Properties		
Property	$x_n = \int_1 X(\phi) e^{j2\pi\phi n} d\phi$	$X(\phi) = \sum_{n=-\infty}^{\infty} x_n e^{-j2\pi\phi n}$
Linearity	$Ax_1[n] + Bx_2[n]$	$\Longleftrightarrow AX_1(\phi) + BX_2(\phi)$
Scale Change (insert $N-1$ zeros)	$\sum_{m=-\infty}^{\infty} x_m \delta_{n-mN}$	$\Longleftrightarrow X(N\phi)$
Time Shift	x_{n-n_0}	$\Longleftrightarrow e^{-j2\pi\phi n_0} X(\phi)$
Frequency Shift	$e^{j2\pi\phi_0 n} x_n$	$\Longleftrightarrow X(\phi - \phi_0)$
Convolution	$\sum_{m=-\infty}^{\infty} x_m h_{n-m}$	$\Longleftrightarrow X(\phi) H(\phi)$
Multiplication	$v_n w_n$	$\Longleftrightarrow \int_1 V(\theta) W(\phi - \theta) d\theta$
Times n	$n x_n$	$\Longleftrightarrow \frac{-1}{j2\pi} \frac{dX(\phi)}{d\phi}$
Summation	$\sum_{m=-\infty}^n x_m$	$\Longleftrightarrow \frac{X(\phi)}{1 - e^{-j2\pi\phi}} + \frac{X(0)}{2} \sum_{k=-\infty}^{\infty} \delta(\phi - k)$
Even Symmetry	$x_n = x_{-n}$	$\Longleftrightarrow X(\phi) = X(-\phi)$
Odd Symmetry	$x_n = -x_{-n}$	$\Longleftrightarrow X(\phi) = -X(-\phi)$
Real x_n	$x_n = x_n^*$	$\Longleftrightarrow X(\phi) = X^*(-\phi)$ ($ X(\phi) $ even, $\angle X(\phi)$ odd)
Periodic x_n	$\frac{1}{N} \sum_{k=0}^{N-1} X_k e^{j2\pi kn/N}$	$\Longleftrightarrow \frac{1}{N} \sum_{k=0}^{N-1} X_k \sum_{m=-\infty}^{\infty} \delta(\phi - \frac{k}{N} - m)$
Parseval	$\sum_{n=-\infty}^{\infty} x_n ^2$	$= \int_1 X(\phi) ^2 d\phi$

Trigonometric Identities

$$e^{\pm j\theta} = \cos \theta \pm j \sin \theta$$

$$\cos \theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta}) = \sin(\theta + 90^\circ)$$

$$\sin \theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta}) = \cos(\theta - 90^\circ)$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\cos^2 \theta - \sin^2 \theta = \cos 2\theta$$

$$\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$$

$$\cos^3 \theta = \frac{1}{4}(3 \cos \theta + \cos 3\theta)$$

$$\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$$

$$\sin^3 \theta = \frac{1}{4}(3 \sin \theta - \sin 3\theta)$$

$$\tan\left(\frac{\alpha}{2}\right) = \sqrt{\frac{1 - \cos \alpha}{1 + \cos \alpha}} = \frac{\sin \alpha}{1 + \cos \alpha} = \frac{1 - \cos \alpha}{\sin \alpha}$$

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

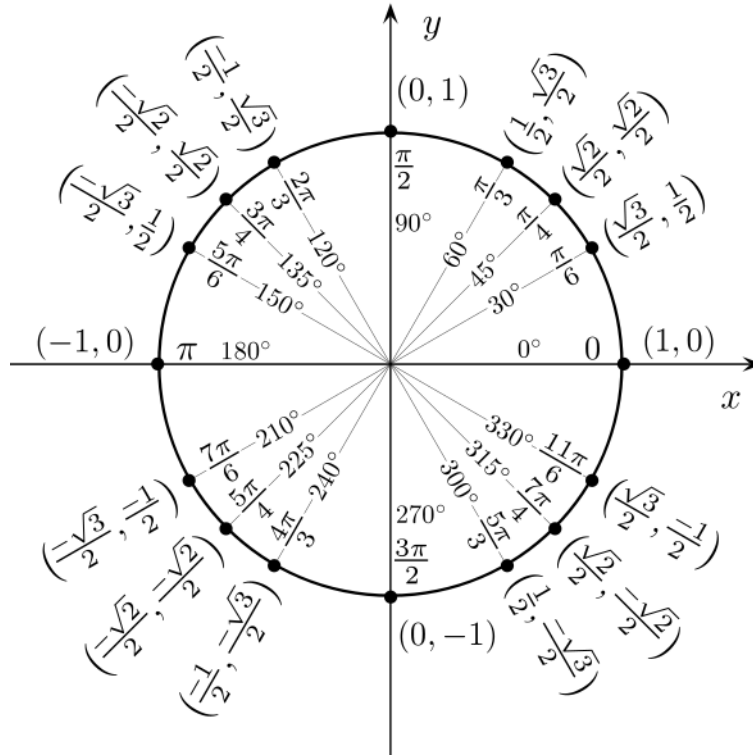
$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

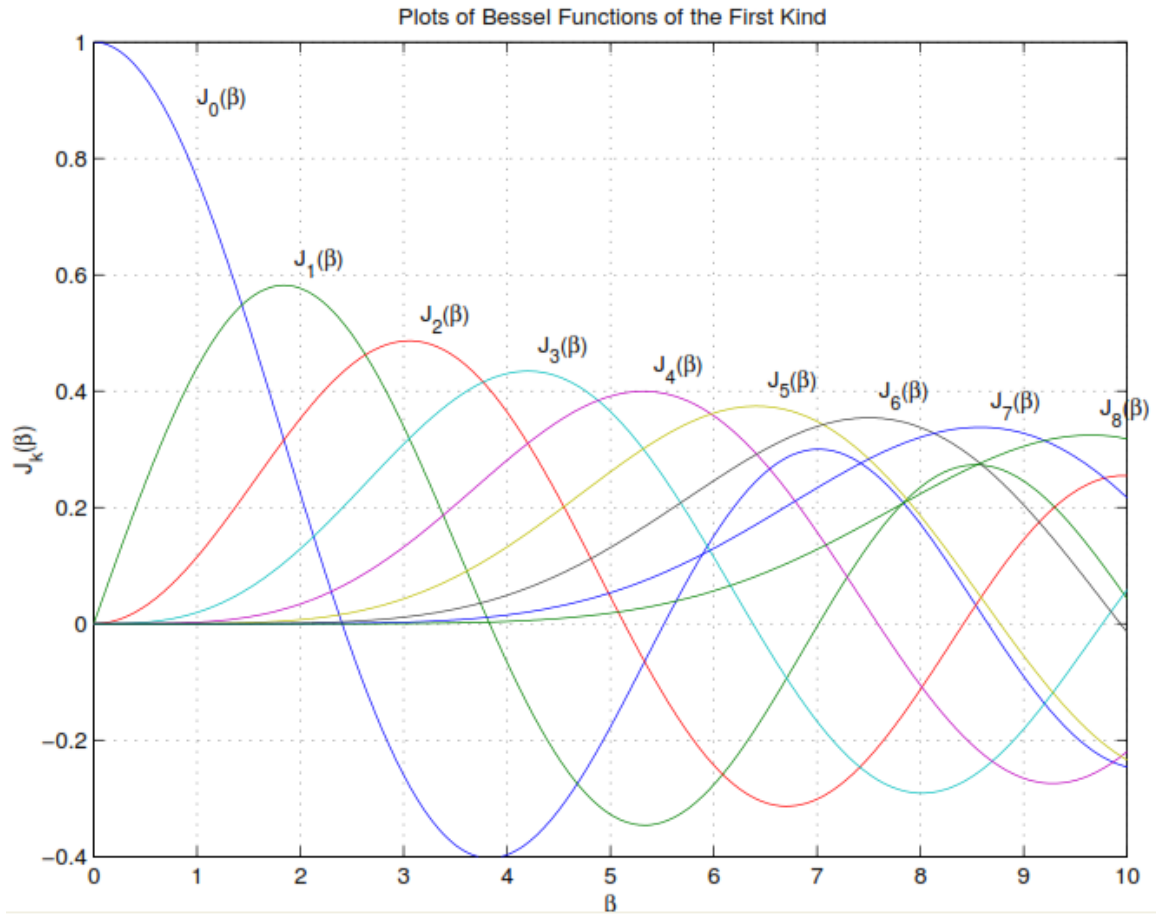
$$\sin \alpha \sin \beta = \frac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$\cos \alpha \cos \beta = \frac{1}{2}[\cos(\alpha - \beta) + \cos(\alpha + \beta)]$$

$$\sin \alpha \cos \beta = \frac{1}{2}[\sin(\alpha - \beta) + \sin(\alpha + \beta)]$$



Bessel Functions



$$J_k(\beta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j(\beta \sin x - kx)} dx, \text{ Bessel function of 1'st kind, order } k.$$

Properties:

1. $J_k(\beta) = J_k^*(\beta)$
2. $J_{-k}(\beta) = (-1)^k J_k(\beta)$
3. $\sum_{k=-\infty}^{\infty} J_k^2(\beta) = 1$

ASCII Code

7-Bit ASCII (American Standard Code for Information Interchange)								
	000...	001...	010...	011...	100...	101...	110...	111...
..0000	<i>NUL</i>	<i>DLE</i>	<i>SP</i>	0	@	P	'	p
..0001	<i>SOH</i>	<i>DC1</i>	!	1	A	Q	a	q
..0010	<i>STX</i>	<i>DC2</i>	"	2	B	R	b	r
..0011	<i>ETX</i>	<i>DC3</i>	#	3	C	S	c	s
..0100	<i>EOT</i>	<i>DC4</i>	\$	4	D	T	d	t
..0101	<i>ENQ</i>	<i>NAK</i>	%	5	E	U	e	u
..0110	<i>ACK</i>	<i>SYN</i>	&	6	F	V	f	v
..0111	<i>BEL</i>	<i>ETB</i>	,	7	G	W	g	w
..1000	<i>BS</i>	<i>CAN</i>	(	8	H	X	h	x
..1001	<i>HT</i>	<i>EM</i>	)	9	I	Y	i	y
..1010	<i>LF</i>	<i>SUB</i>	*	:	J	Z	j	z
..1011	<i>VT</i>	<i>ESC</i>	+	;	K	[	k	{
..1100	<i>FF</i>	<i>FS</i>	,	<	L	\	l	
..1101	<i>CR</i>	<i>GS</i>	-	=	M	]	m	}
..1110	<i>SO</i>	<i>RS</i>	.	>	N	^	n	~
..1111	<i>SI</i>	<i>US</i>	/	?	O	_	o	<i>DEL</i>

Q Function

Q(x) for x = arange(1,8,0.1)

	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9
1.	1.587e-01	1.357e-01	1.151e-01	9.680e-02	8.076e-02	6.681e-02	5.480e-02	4.457e-02	3.593e-02	2.872e-02
2.	2.275e-02	1.786e-02	1.390e-02	1.072e-02	8.198e-03	6.210e-03	4.661e-03	3.467e-03	2.555e-03	1.866e-03
3.	1.350e-03	9.676e-04	6.871e-04	4.834e-04	3.369e-04	2.326e-04	1.591e-04	1.078e-04	7.235e-05	4.810e-05
4.	3.167e-05	2.066e-05	1.335e-05	8.540e-06	5.413e-06	3.398e-06	2.112e-06	1.301e-06	7.933e-07	4.792e-07
5.	2.867e-07	1.698e-07	9.964e-08	5.790e-08	3.332e-08	1.899e-08	1.072e-08	5.990e-09	3.316e-09	1.818e-09
6.	9.866e-10	5.303e-10	2.823e-10	1.488e-10	7.769e-11	4.016e-11	2.056e-11	1.042e-11	5.231e-12	2.600e-12
7.	1.280e-12	6.238e-13	3.010e-13	1.439e-13	6.811e-14	3.192e-14	1.482e-14	6.828e-15	3.109e-15	1.388e-15